

Problem set 3
Due Thursday September 24, 11:59pm

Exercise 3.1 (5 points). Prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

Exercise 3.2 (5 points). Let X be any set, and let $d : X \times X \rightarrow \mathbb{R}$ be the discrete metric, defined by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

for all $x, y \in X$.

- (1) Prove that, with this distance function, X is a metric space.
- (2) For any $x \in X$, what is the neighbourhood $N_\epsilon(x)$ when $\epsilon = \frac{1}{2}$, 1, and 2?
- (3) Which subsets of X are open? Which are closed?

Exercise 3.3 (5 points). Show that the subset of $\mathbb{R}^2$ given by

$$E = \{(x, y) \in \mathbb{R}^2 \mid x < y\}$$

is open.

Exercise 3.4 (10 points; Rudin 2.2+2.3, modified). A real number x is called *algebraic* if there exist $n \in \mathbb{N}$ and $a_0, \dots, a_n \in \mathbb{Z}$, with $a_0 \neq 0$, such that

$$a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n = 0.$$

- (1) Prove that $\sqrt{5}$ and $\sqrt{2 + \sqrt{3}}$ are algebraic.
- (2) Prove that the set of all algebraic real numbers is countable. (You may use without proof the fact that a polynomial of degree n has at most n roots.)
- (3) Prove that there exist real numbers which are not algebraic.

Exercise 3.5 (10 points). In class, we will see that a sequence of bounded closed intervals $E_1, E_2, E_3, \dots$ with $E_{i+1} \subset E_i$ always has non-empty intersection. In this exercise we see that this is a very special property.

- (1) Find a sequence of open intervals $U_1, U_2, U_3, \dots$ in $\mathbb{R}$ with $U_{k+1} \subset U_k$ such that the intersection $\bigcap_{k \in \mathbb{N}} U_k$ is empty.
- (2) Find a sequence of closed sets $C_1, C_2, C_3, \dots$ in $\mathbb{R}$ with $C_{k+1} \subset C_k$ such that the intersection $\bigcap_{k \in \mathbb{N}} C_k$ is empty.
- (3) Find a sequence of dense sets $D_1, D_2, D_3, \dots$ in $\mathbb{R}$ with $D_{k+1} \subset D_k$ such that the intersection $\bigcap_{k \in \mathbb{N}} D_k$ is empty. Prove the D_i you define are dense.